

**Δ vs frozen (good-oriented, value + 95% CI) · PRETRAIN block / SURGERY block · SCORE = mean per-metric-normalized (0-1) · BLUE BOX + big Δ = best arm per metric
WINNER row (champion duel — S=surgery P=pretrain ==tie · tie = paired 95% CI overlaps 0): surgery 4 · pretrain 0 · tie 5 → SURGERY**

	action top1 ↑ better classify motion class from frozen features	motion_cos ↑ better same-motion clips cluster vs different motions	taxonomy f1 ↑ better read scene attributes from features	future mse ↓ better predictor guesses masked next-frame tokens	rollout ↓ better error growth over multi-step rollout	causal ↓ better predict 2nd half from 1st half	tdist ↓ better accuracy decay vs prediction horizon	teacher_free ↓ better free-run minus teacher-forced gap	maskratio ↓ better degradation under heavier masking	SCORE (norm 0-1)
ViT-G pretrain_2x encoder	+8.77 [+6.47, +11.01]	+0.16 [+0.16, +0.16]	-0.01 [-0.02, +0.01]	+0.01 [+0.01, +0.01]	0 [0, 0]	+0.01 [+0.01, +0.01]	0 [0, 0]	0 [0, 0]	0 [0, 0]	0.44
ViT-G pretrain encoder	+7.29 [+4.99, +9.48]	+0.15 [+0.15, +0.15]	0 [-0.02, +0.01]	+0.02 [+0.02, +0.02]	0 [0, 0]	+0.02 [+0.02, +0.02]	0 [0, 0]	0 [0, 0]	0 [0, 0]	0.74
ViT-G pretrain head	-1.53 [-3.65, +0.49]	0 [0, 0]	0 [-0.01, +0.01]	0 [0, 0]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0.35
ViT-G surgery 3stage_DI encoder	+5.92 [+3.62, +8.22]	+0.15 [+0.15, +0.16]	0 [-0.02, +0.02]	+0.04 [+0.04, +0.04]	0 [0, 0]	+0.04 [+0.04, +0.04]	0 [0, 0]	0 [0, 0]	+0.01 [+0.01, +0.01]	0.67
ViT-G surgery 3stage_DI head	+6.85 [+4.55, +9.15]	+0.16 [+0.15, +0.16]	0 [-0.02, +0.01]	+0.02 [+0.02, +0.02]	0 [0, 0]	+0.02 [+0.02, +0.02]	0 [0, 0]	0 [0, 0]	0 [0, 0]	0.71
ViT-G surgery noDI encoder	+7.40 [+4.93, +9.64]	+0.16 [+0.15, +0.16]	0 [-0.02, +0.01]	+0.04 [+0.04, +0.04]	0 [0, 0]	+0.04 [+0.04, +0.04]	0 [0, 0]	0 [0, 0]	+0.01 [+0.01, +0.01]	0.61
ViT-G surgery noDI head	+7.34 [+4.99, +9.64]	+0.17 [+0.17, +0.17]	-0.01 [-0.02, +0.01]	+0.02 [+0.02, +0.02]	0 [0, 0]	+0.02 [+0.02, +0.02]	0 [0, 0]	0 [0, 0]	0 [0, 0]	0.71
WINNER	=	S	=	S	=	S	=	=	S	